DAY 03

NETWORK INTERFACES AND ETHERNET

Practical Report

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Topic: Network Interface Cards, Ethernet, and Network Configuration

PART A:

1. What is a Network Interface Card (NIC)?

A NIC is a hardware component that enables a computer to connect to a network and communicate with other devices.

2. What is Ethernet?

Ethernet is a widely used networking technology for connecting devices in a Local Area Network (LAN), usually through wired connections.

3. What is RJ-45?

RJ-45 is the commonly used connector for Ethernet cables, connecting network cables to computers, routers, switches, and other devices.

4. What does UTP stand for?

UTP stands for Unshielded Twisted Pair, a cable made of twisted copper-wire pairs without additional metallic shielding.

5. Name three common Ethernet cable categories.

Cat5e, Cat6, and Cat6a.

6. What is a straight-through cable?

A straight-through cable uses the same wiring standard at both ends: T568A–T568A or T568B–T568B. It commonly connects different types of network devices.

7. What is a crossover cable?

A crossover cable uses T568A at one end and T568B at the other. Traditionally, it connected similar devices directly, such as two computers.

8. Explain Auto MDI-X.

Auto MDI-X automatically detects and adjusts Ethernet transmit and receive pairs, reducing the need for crossover cables.

9. Difference between copper and fiber-optic networking?

Copper carries data as electrical signals through copper wires. Fiber carries data as light through optical fibers and generally supports longer distances, high bandwidth, and immunity to electromagnetic interference.

10. What is multimode fiber?

Multimode fiber has a relatively large core that allows multiple light paths. It is commonly used for short-distance links in buildings and data centers.

11. What is single-mode fiber?

Single-mode fiber has a small core that supports essentially one propagation mode and is suitable for high-bandwidth, long-distance communication.

12. Two advantages of fiber-optic networking.

• High bandwidth over long distances.
• Immunity to electromagnetic interference.

13. Why is understanding network interfaces important for cybersecurity?

It helps security professionals identify active connections, inspect IP addresses and gateways, troubleshoot network issues, monitor traffic, and investigate suspicious connections.

PART B: PRACTICAL IDENTIFICATION

The active adapter shown in the screenshot is **Wireless LAN adapter Wi-Fi**. The computer also shows an **Ethernet adapter Ethernet 2** with a separate IPv4 address; confirm which connection you are using if your instructor requires only one active interface.

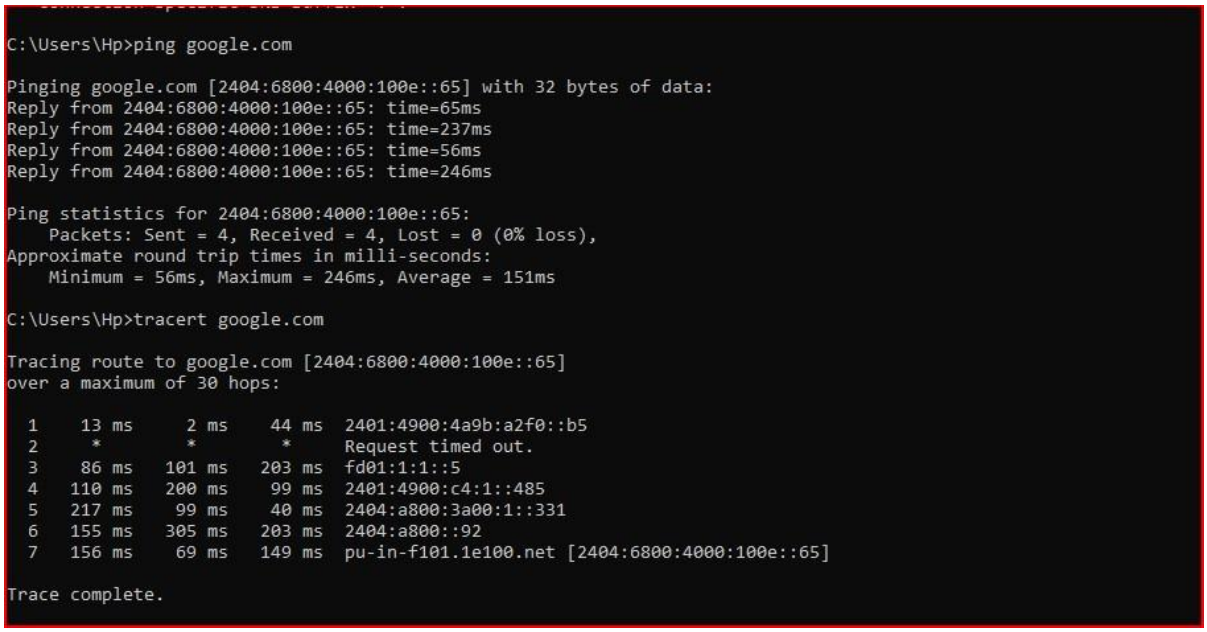
Network configuration

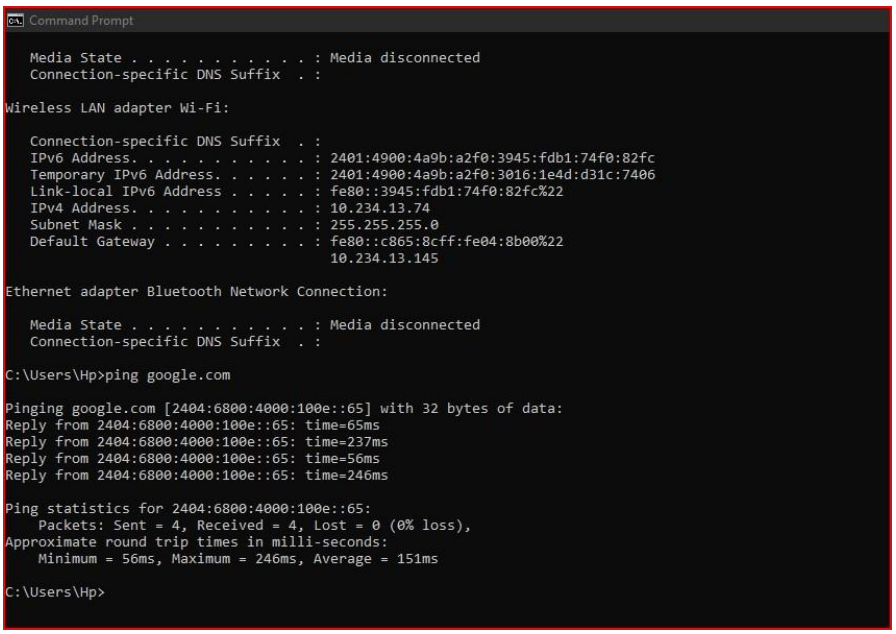
Field	Value from screenshot
Network adapter (Wi-Fi)	Wireless LAN adapter Wi-Fi
IPv4 address (Wi-Fi)	10.234.13.74
Default gateway (Wi-Fi)	10.234.13.145
Subnet mask (Wi-Fi)	255.255.255.0
Additional adapter	Ethernet adapter Ethernet 2
IPv4 address (Ethernet 2)	192.168.56.1
Default gateway (Ethernet 2)	Not shown

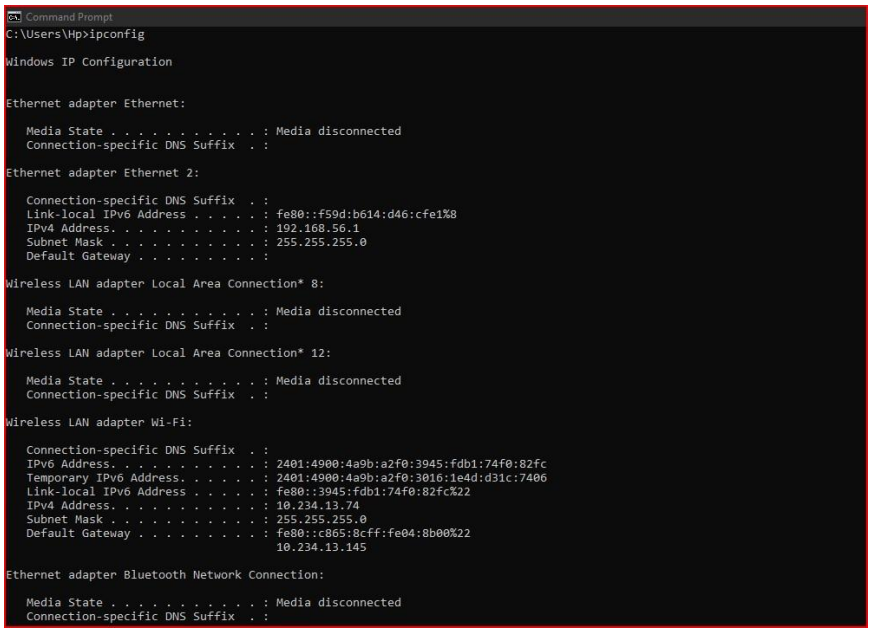
Note: The Wi-Fi screenshot shows an IPv4 default gateway of 10.234.13.145. The other Ethernet 2 interface shows 192.168.56.1 and no default gateway.

PART C: SCREENSHOTS

The screenshots supplied are included below as practical evidence. The first shows network connectivity tests, the second shows Wi-Fi configuration and ping results, and the third shows the ipconfig output.




Command Prompt
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2401:4900:4a9b:a2f0:3945:fdb1:74f0:82fc
Temporary IPv6 Address. . . . . : 2401:4900:4a9b:a2f0:3016:1e4d:d31c:7406
Link-local IPv6 Address . . . . . : fe80::3945:fdb1:74f0:82fc%22
IPv4 Address. . . . . : 10.234.13.74
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::c865:8cff:fe04:8b00%22
10.234.13.145
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\Hp>ping google.com
Pinging google.com [2404:6800:4000:100e::65] with 32 bytes of data:
Reply from 2404:6800:4000:100e::65: time=65ms
Reply from 2404:6800:4000:100e::65: time=237ms
Reply from 2404:6800:4000:100e::65: time=56ms
Reply from 2404:6800:4000:100e::65: time=246ms
Ping statistics for 2404:6800:4000:100e::65:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 56ms, Maximum = 246ms, Average = 151ms
C:\Users\Hp>


Command Prompt
C:\Users\Hp>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Ethernet adapter Ethernet 2:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::f59d:b614:d46:cfe1%8
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Wireless LAN adapter Local Area Connection* 8:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 12:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2401:4900:4a9b:a2f0:3945:fdb1:74f0:82fc
Temporary IPv6 Address. . . . . : 2401:4900:4a9b:a2f0:3016:1e4d:d31c:7406
Link-local IPv6 Address . . . . . : fe80::3945:fdb1:74f0:82fc%22
IPv4 Address. . . . . : 10.234.13.74
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::c865:8cff:fe04:8b00%22
10.234.13.145
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :